

Method	Animal-Animal			Animal-Object			Object-Object		
	Full Sim.	Min. Sim.	T-C Sim.	Full Sim.	Min. Sim.	T-C Sim.	Full Sim.	Min. Sim.	T-C Sim.
SD [35]	0.311	0.213	0.767	0.340	0.246	0.793	0.335	0.235	0.765
CD [24]	0.284	0.232	0.692	0.336	0.252	0.769	0.349	0.265	0.759
StrD [11]	0.306	0.210	0.761	0.336	0.242	0.781	0.332	0.234	0.762
EBCA [29]	0.291	0.215	0.722	0.317	0.229	0.732	0.321	0.231	0.726
AnE [6]	0.332	0.248	0.806	0.353	0.265	0.830	0.360	0.270	0.811
SG [32]	0.311	0.213	0.767	0.355	0.264	0.830	0.355	0.262	0.811
EBAMA [44]	0.340	0.256	0.817	0.362	0.270	0.851	0.366	0.274	0.836
D&B [22]	0.331	0.246	0.810	—	—	—	—	—	—
Ours	0.348	0.262	0.823	0.370	0.276	0.868	0.370	0.275	0.844

Table 1. **Comparison of Full Sim., Min. Sim., and T-C Sim. across different methods on the AnE dataset. CLIP Score based on the Attend and Excite Dataset.** Note that the performance of SG on Animal-Animal is degraded to SD, as the prompts do not contain any attribute-object pairs. The best results for each category is marked in bold numbers.

$\mathcal{O} = \{1, 2, \dots, n\}$, define $\mathcal{S} = \bigcup_{k \in \mathcal{O}} (\mathcal{M}_k \cup \mathcal{N}_k)$ as the set of all indices associated with object-related tokens. The set of outside tokens, $\mathcal{O}_{\text{outside}}$, includes all other token indices not in $\mathcal{S}$.

The outside loss is defined as:

$$\mathcal{L}_{\text{out}} = -\frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \left(\max_{j \in \mathcal{O}_{\text{out}}} D_{\text{sym}}(A_i, A_j) \right) \quad (11)$$

This loss penalizes high similarity between object-specific tokens and unrelated tokens, thereby encouraging the model to focus attention on relevant parts of the prompt and improving the clarity and accuracy of the generated images.

4.4.4 PAC-Bayes Regularizer

The PAC-Bayes regularizer $\mathcal{R}_{\text{PAC}}$ applies principles from PAC-Bayes theory to regularize the attention distributions, promoting generalization and preventing overfitting. For a given set of samples, let N be the number of samples and δ be the confidence parameter. The regularizer is defined as:

$$\mathcal{R}_{\text{PAC}} = -\sqrt{\frac{D_{\text{KL}}(A \parallel U) + \log\left(\frac{2\sqrt{N}}{\delta}\right)}{2N}} \quad (12)$$

where $D_{\text{KL}}(A \parallel U)$ represents the KL divergence between the learned attention distribution A and the uniform distribution U over the attention map dimensions.

The PAC-Bayes regularizer guides the attention map away from uniformity and encourages the attention map to focus on more informative patterns. This approach helps the model generalize by emphasizing meaningful structures in the data, while still avoiding overfitting.

5. Workflow

Our method integrates custom attention priors into the denoising process of text-to-image diffusion models, specifically focusing on Stable Diffusion [35]. We start by parsing

the input prompt with a natural language parser to extract nouns and their associated modifiers, identifying relationships among them. In each denoising step, we capture the cross-attention maps for these tokens and apply custom loss functions guided by the PAC-Bayes framework [25]. These loss functions, based on the Kullback-Leibler divergence between the learned attention distributions and our designed priors, encourage divergence between different objects’ attention maps, ensure similarity between modifiers and their nouns, minimize attention to irrelevant tokens, and include a regularization term to prevent overfitting. Together, these losses iteratively update the latent representations, refining the generated image to align more closely with the prompt and enhancing attribute binding accuracy, as outlined in Algorithm 1.

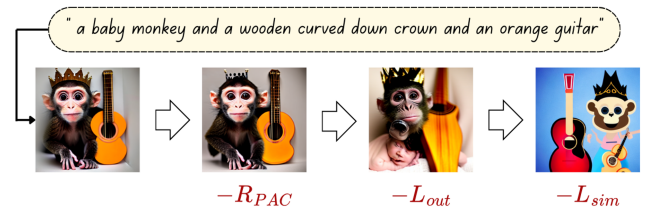


Figure 5. **Qualitative Results of the Ablation Study.** Each column illustrates the generated image for the prompt ‘a baby monkey and a wooden curved crown and an orange guitar,’ showing the effects of omitting specific loss components. From left to right: the full model output, the output without the PAC Regularizer, the output without the Outside Loss, and the output without the Similarity Loss. Each ablation demonstrates the impact on attribute-object alignment and overall image coherence, with attribute misbindings and inconsistencies appearing as each component is removed.

6. Experiment

Evaluation Datasets We evaluate the effectiveness of our proposed method on three benchmark datasets: Attend-and-Excite (AnE) [6], ABC-6K, and DVMP [32]. The AnE